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Project 2

**Structural Determinants of Wealth: A
Global Analysis of GDP per Capita (2011–
2021)**

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1- Abstract

This report investigates the structural determinants of GDP per capita across countries, focusing on the influence of factors such as industrial activity, innovation, education, demographics, and natural resource dependence. The analysis is based on a harmonized panel dataset constructed from multiple sources, which required extensive data cleaning, normalization, imputation of missing values, and integration of temporal and spatial identifiers.

We employ a combination of statistical and machine learning techniques, including OLS regression with cluster-robust standard errors, Random Forest importance ranking, and unsupervised clustering using K-Means. The results reveal that manufacturing share, innovation index, and tertiary education enrolment are strong positive predictors of economic development, while resource rents show mixed effects depending on the model used. Clustering analysis further uncovers distinct economic profiles among countries, based on structural and institutional variables.

This multi-method approach provides valuable insights for understanding the economic trajectories of nations and supports evidence-based policy design aimed at promoting inclusive and sustainable growth.

2- Introduction

Explaining why some nations enjoy income levels many times higher than others is a classic challenge in economics—and an ideal case study for a Systems and Data Management (SGD) course. Public analyses often look at GDP per capita through a single lens—education or natural resources—yielding piecemeal insights.

This project assembles, for more than 128 economies over 2011-2021, a harmonised panel that merges seven officially curated indicators:

- GDP per capita (current US\$)
- Manufacturing value added (% of GDP)
- Natural-resource rents (% of GDP)
- Tertiary-education enrolment (% gross)
- Innovation capacity (Global Innovation Index score)
- Population density (people/km²)
- Working-age share (population aged 15-64, %)

With this dataset we address three guiding questions:

1. Which structural factors explain most of the cross-country variance in GDP per capita?
2. Do innovation and higher education outweigh natural resources once industrial structure is controlled for?
3. Can countries be clustered into distinct economic profiles based on these drivers?

In addition, several secondary questions emerge through the modelling process:

- What is the relative importance of innovation, education, and resource rents when assessed jointly?
- Do structural drivers have different effects in low-income versus high-income contexts?
- Is there evidence of interaction or substitution between innovation and education?
- How relevant are demographic and spatial factors such as population density or working-age share?
- Do non-linear models like Random Forest outperform traditional regression in predictive power?

The study was chosen because it:

- Aligns perfectly with SGD goals—collecting heterogeneous data, normalising formats, storing everything in a single SQLite database and running end-to-end analytics;
- Meets Project 2 – Option 1 requirements—we reproduce official indicators and extend them with additional sources to deliver new evidence;
- Provides actionable value—findings can inform policy on industrial upgrading, R&D incentives and education investment.

The remainder of the report is organised as follows: Section 3 details data sources; Section 4 describes preparation and integration; Section 5 presents exploratory analysis; Section 7 outlines the modelling approach; Section 8 discusses results; Section 9 covers limitations and future work; Section 10 concludes.

3- Data Sources

This project draws upon two main open-access data repositories: the World Bank DataBank and the World Intellectual Property Organization (WIPO) via Kaggle. Together, these sources provide complementary structural indicators for over 150 countries between 2011 and 2021, enabling robust cross-country panel analysis.

3.1 World Bank DataBank

The World Bank was the primary source for six out of the seven indicators used. Data were retrieved from the [World Bank DataBank](#) by exporting separate CSV files corresponding to each of the following series:

- GDP per capita (current US\$) – an indicator of national income per person;
- Manufacturing value added (% of GDP) – representing industrialisation level;
- Natural-resource rents (% of GDP) – reflecting reliance on extractive resources;
- Tertiary school enrolment (% gross) – a proxy for access to higher education;
- Population density (people per km²) – capturing spatial population concentration;
- Working-age population (% aged 15–64) – a key demographic driver of productivity.

Each file contained data for the years 2011 to 2021, across 217 countries and territories. The structure of each file followed a consistent format, with the columns:

- Series Name, Series Code, Country Name, Country Code
- Annual values from 2011 to 2021, formatted as 2011 [YR2011], 2012 [YR2012], ..., 2021 [YR2021].

These files were later cleaned, reshaped, and merged into a unified panel using consistent country identifiers and year fields.

3.2 Global Innovation Index (WIPO via Kaggle)

Innovation capacity was captured using the Global Innovation Index (GII), sourced from a curated Kaggle dataset:

<https://www.kaggle.com/datasets/karlovakacs/global-innovation-index-wipo-2011-2024-data>

From this dataset, the file `GII_2011_2024_long_format.tsv` was extracted. It contains country-level innovation scores from 2011 to 2024, although only data for 2011–2021 were used for consistency. The file has the following columns:

- country – the three-letter country code following the ISO;
- year – year of the observation;
- rank – ordinal rank in that year's GI;
- score – GI score on a scale from 0 to 100, used in this project as the innovation proxy.

Only the rank column was discarded; all others were retained and later used during data preparation. In particular, the country (ISO code) and year fields played a key role in aligning this source with the World Bank data, ensuring consistent merging across datasets.

4 - Data Preparation & Integration

The raw data obtained from Kaggle and the World Bank required substantial cleaning and transformation to ensure consistency across sources and readiness for analysis. This process was divided into two main preprocessing scripts: `transform_gii.py` for innovation data and `transform_worldbank_files.py` for the remaining variables.

4.1 Global Innovation Index Cleanup

The Global Innovation Index (GII) data was stored in a tab-separated file (`GII_score.tsv`). The cleaning steps performed in `transform_gii.py` were as follows:

- Column filtering: Only three fields were retained: Country, Year, and Score (the GII score).
- Renaming: These columns were renamed to `country`, `year`, and `gii_score` to standardise naming conventions across datasets.
- Missing values: A score of -1, used in the original file to indicate missing data, was replaced with NaN.
- Export: The cleaned dataset was saved as a CSV file named `GII_score_long.csv`.

This yielded a long-format dataset ready to be merged with other sources using the common keys: country (ISO 3-letter code) and year.

4.2 World Bank Data Cleanup

The World Bank indicators were initially provided as separate CSV files, each containing multiple header rows and data in wide format (columns for each year from 2011 to 2021).

A preprocessing script (`transform_worldbank_files.py`) was used to automate the transformation of each file into a standard format:

- Header and footer removal: The first four metadata lines and six footer lines were skipped using `skipfooter=6`.
- Year selection: Only columns corresponding to 2011–2021 were retained.
- Reshaping: Data were converted from wide to long format using `pandas.melt`, producing one row per country-year observation.
- Year extraction: The year label (e.g. "2011 [YR2011]") was parsed to extract the numeric year.
- Renaming and export: The indicator column was renamed (e.g., `age1564_pct`, `population_density`, etc.), and the data were sorted by country and year before being saved.

This transformation pipeline was applied to six indicators, generating the following harmonized files:

- `gdp_per_capita_long.csv`
- `manufacturing_pct_gdp_long.csv`
- `resource_rents_pct_gdp_long.csv`
- `tertiary_enrolment_pct_long.csv`
- `population_density_long.csv`

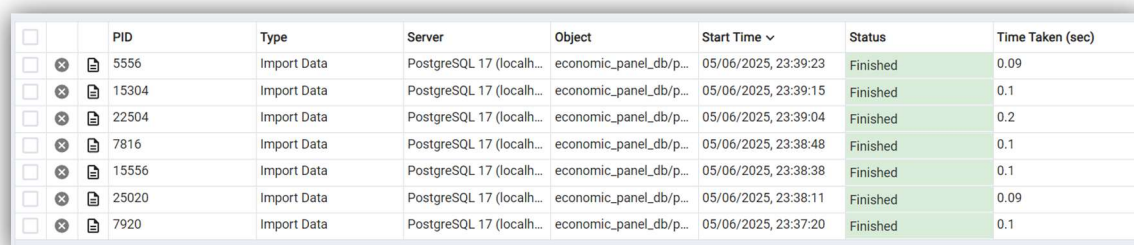
- age1564_pct_long.csv

All files follow a consistent structure with columns: country, year, and value, and were ready for integration with the GII dataset.

4.3 Merge of datasets

After cleaning and transforming the original datasets into tidy, long-format CSV files, the next step was to import the data into a structured SQL database to support merging and querying efficiently.

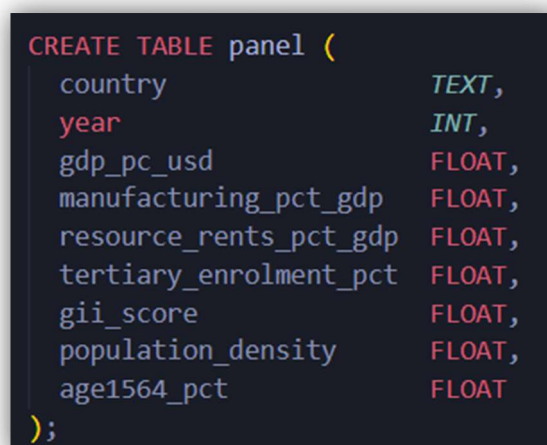
A new PostgreSQL database named `economic_panel` was created in pgAdmin. Within it, seven tables were defined—one for each indicator—alongside an empty panel table to store the merged dataset. The table creation was executed using the script `01_create_tables.sql` that goes with the delivery.



	PID	Type	Server	Object	Start Time	Status	Time Taken (sec)
☐	5556	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:39:23	Finished	0.09
☐	15304	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:39:15	Finished	0.1
☐	22504	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:39:04	Finished	0.2
☐	7816	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:38:48	Finished	0.1
☐	15556	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:38:38	Finished	0.1
☐	25020	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:38:11	Finished	0.09
☐	7920	Import Data	PostgreSQL 17 (localh...	economic_panel_db/p...	05/06/2025, 23:37:20	Finished	0.1

Figure 1. Log of import operations into the PostgreSQL database (`economic_panel_db`).

Each table used only three columns: country (ISO3 code), year, and the corresponding metric value. Additionally, an empty table named `panel` was declared to later receive the full join of all indicators.



```
CREATE TABLE panel (
  country          TEXT,
  year             INT,
  gdp_pc_usd       FLOAT,
  manufacturing_pct_gdp  FLOAT,
  resource_rents_pct_gdp  FLOAT,
  tertiary_enrolment_pct  FLOAT,
  gii_score        FLOAT,
  population_density  FLOAT,
  age1564_pct      FLOAT
);
```

Figure 2. SQL schema used to create the `panel` table, containing the seven variables included in the final model: GDP per capita, manufacturing share, resource rents, tertiary enrolment, innovation, population density, and age structure.

All cleaned `.csv` files were then imported directly into their corresponding tables using pgAdmin's CSV import tool. This setup allowed SQL-based joins to be used in the next step for efficient merging and handling of missing data.

After populating the database with all cleaned tables, a second SQL script (02_build_full_panel.sql) was executed to join all variables into a unified panel. This script uses the gdp table that contains the data from gdp_per_capita_long.csv as the base reference—ensuring that each country-year with GDP data from 2011 to 2021 is included—and performs left joins with all remaining indicator tables.

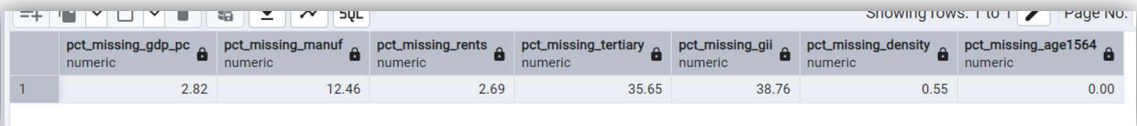
This logic guarantees that all country-year pairs available in GDP data are preserved, even if some indicators are missing for certain years. The result is a panel with partial observations (i.e., NULLs in some columns), which ensures maximum temporal and spatial coverage before further cut and imputation steps.

4.4 Filtering and cleaning for future analysis and modelling

Before proceeding with any modelling, we assessed the completeness of the unified panel using the SQL script 03_check_coverage_and_missing.sql. This script computed the total number of country-year observations and the proportion of missing values (NULL) for each indicator column in the panel table.

The results showed that while most indicators were well covered, some variables exhibited significant gaps.

These figures indicate that GII scores and tertiary enrolment rates were the most incomplete, as expected because GII only covered 150 countries and the others due to unreported data for certain countries or years. By contrast, demographic and structural economic indicators had near-complete coverage.



	pct_missing_gdp_pc numeric	pct_missing_manuf numeric	pct_missing_rents numeric	pct_missing_tertiary numeric	pct_missing_gii numeric	pct_missing_density numeric	pct_missing_age1564 numeric
1	2.82	12.46	2.69	35.65	38.76	0.55	0.00

Figure 3. Percentage of missing values per variable.

This analysis was essential to inform the choice of imputation and deletion strategies.

To prepare the dataset for analysis and modelling, we applied a filtering step to retain only country-year observations where both the target variable (GDP per capita) and the key explanatory variable (innovation score, via GII) were available. This step was implemented using the script 04_build_panel_model.sql, which generated a new table named panel_model.

This operation had two goals:

- Ensure data consistency by keeping only entries where the dependent variable (gdp_pc_usd) is known;
- Preserve the study’s core analytical theme by requiring a valid gii_score, the main proxy for innovation capacity.

Out of the full panel, 1375 records remained after this filtering step. We then assessed the remaining missing data in the explanatory variables:

	missing_manuf_after bigint	missing_rents_after bigint	missing_tertiary_after bigint	missing_density_after bigint	missing_age1564_after bigint
1	39	0	226	1	0

Figure 4. Remaining number of missing values.

This confirmed that although the panel was largely complete, some degree of imputation would be necessary—especially for tertiary education indicators.

In addition to quantifying missing values, we identified which countries had entirely missing data for specific variables—i.e., no available values across any year. This was useful to assess whether certain countries might introduce irrecoverable gaps in the panel. The query `05_list_all-null_countries.sql` revealed:

	missing_manuf text	missing_tertiary text	missing_density text
1	BGR	BOL	[null]
2	SDN	BRB	[null]
3	SYR	BTN	[null]
4	YEM	GAB	[null]
5	[null]	MMR	[null]
6	[null]	PRY	[null]
7	[null]	TTO	[null]
8	[null]	VEN	[null]

Figure 5. Sample of countries with missing data for key variables.

Countries that had no valid values at all for any of the three variables across the entire 2011–2021 period were removed.

This operation was implemented in the script `06_remove_all-null_countries.sql` and followed a prior diagnostic query (`05_list_all-null_countries.sql`) that identified countries like *BGR*, *SDN*, *SYR*, *VEN* and others as having all observations missing for at least one of these indicators.

These countries were excluded because they offered no usable data for some of the most critical explanatory variables, and their presence would distort imputation and analysis. After this removal, the resulting table `panel_model` retained all countries with at least one valid entry per variable, making it a coherent base for statistical modelling.

4.5 – Imputation of Missing Values

After exporting the cleaned and pruned `panel_model` table from PostgreSQL, a final preprocessing step was performed using the Python script `panel_model_imputation.py`. This script handled the remaining missing values in the `panel_model`.

The imputation was performed by filling missing values with the mean of each country for the corresponding indicator. This method ensured that imputations respected the national context, using only data from the same country to estimate missing entries.

After this step, both variables had no remaining missing values, making the dataset suitable for statistical modelling. Additionally, a new variable `log_gdp` was created by computing the natural logarithm of GDP per capita to help stabilise variance and improve model interpretability. The resulting file, `panel_model_final.csv`, was saved for use in subsequent analytical steps.

5 - Exploratory Analysis

5.1 Overview of the Dataset

To understand the structural determinants of GDP per capita, we first performed an exploratory analysis on the final cleaned and imputed panel dataset (panel_model_final.csv). This dataset was processed using the custom Python script main_analysis.py.

To ensure sufficient temporal robustness in the statistical modeling phase, we excluded countries with fewer than 5 years of valid data. This threshold helps ensure that the estimates of time-averaged indicators (e.g., industrial share, education levels) are stable and not overly influenced by short-term fluctuations or missing years. Moreover, several parts of the analysis—such as regression modeling and clustering—implicitly assume that each country's values represent stable long-run trends, which is unrealistic with less than 5 data points.

After filtering, the dataset retained 128 countries (out of 134 initially considered), each with at least 5 valid years of observations, resulting in 1,356 country-year records.

```
=== Estatísticas de cobertura por país (mínimo 5 anos) ===
count      134.000000
mean       10.261194
std         1.992199
min         2.000000
25%         11.000000
50%         11.000000
75%         11.000000
max         11.000000
Name: count, dtype: float64

Países com menos de 5 anos (serão excluídos):
country
LSO      4
GMB      4
FJI      4
LAO      3
SYC      2
BLZ      2
Name: count, dtype: int64

Total de países após filtragem: 128 (de 134)
```

Figure 6. Coverage statistics showing the number of years with data per country. Only countries with at least 5 years of valid data were retained, resulting in a final sample of 128 countries.

5.2 Descriptive Statistics

The table below presents descriptive statistics for the main variables used in the analysis, including the dependent variable `log_gdp` (natural logarithm of GDP per capita in USD) and six explanatory variables related to industry, innovation, education, demographics, and natural resources:

```
=== Descriptive Statistics ===
```

	<code>manufacturing_pct_gdp</code>	<code>resource_rents_pct_gdp</code>	<code>tertiary_enrolment_pct</code>	<code>gii_score</code>	<code>population_density</code>	<code>age1564_pct</code>	<code>log_gdp</code>
count	1356.00	1356.00	1356.00	1356.00	1356.00	1356.00	1356.00
mean	12.96	18241.68	47.10	37.12	276.43	64.66	25.40
std	5.59	22666.23	28.98	11.81	944.20	6.45	1.85
min	0.91	210.24	0.85	15.00	1.79	48.13	21.28
25%	9.21	2952.42	20.40	28.05	44.95	62.18	23.84
50%	12.25	7492.79	48.16	34.80	89.99	65.53	25.20
75%	16.28	26764.10	69.05	45.66	187.27	68.22	26.67
max	34.86	133711.79	150.61	68.40	7965.88	85.21	30.80

Figure 7. Summary statistics for all variables in the final panel.

These statistics highlight a high degree of variability across countries:

- Tertiary education enrollment ranges from under 1% to over 150%, reflecting stark contrasts between developing and developed economies, and possibly the presence of over-aged students or foreign enrollees.
- Resource rents show extreme skewness, with some economies heavily dependent.
- Population density varies from sparse (less than 2 people/km²) to extremely dense (close to 8,000/km²), indicating different spatial and urbanization contexts.
- The Global Innovation Index exhibits substantial dispersion, with some countries scoring below 20 and others above 65, suggesting wide gaps in innovation capacity.

These differences justify the need for standardization before regression modeling and motivate the cluster analysis of country profiles.

5.2 Distribution of GDP per Capita

To better understand the output variable's characteristics, the figure 8 displays the probability density of the GDP per capita (`gdp_pc_usd`) across all country-year observations. The distribution is highly right-skewed, with most countries concentrated at low income levels and a long tail extending toward higher-income countries. This reflects substantial global inequality.

To reduce this skewness and better meet the assumptions of regression models, we transformed the variable using the natural logarithm. Figure 9 shows the resulting distribution of `log_gdp`, which is much closer to a normal distribution, with a multi-modal structure that suggests the existence of distinct economic groups (e.g., low-, middle- and high-income countries). This transformation is used in all subsequent modelling steps.

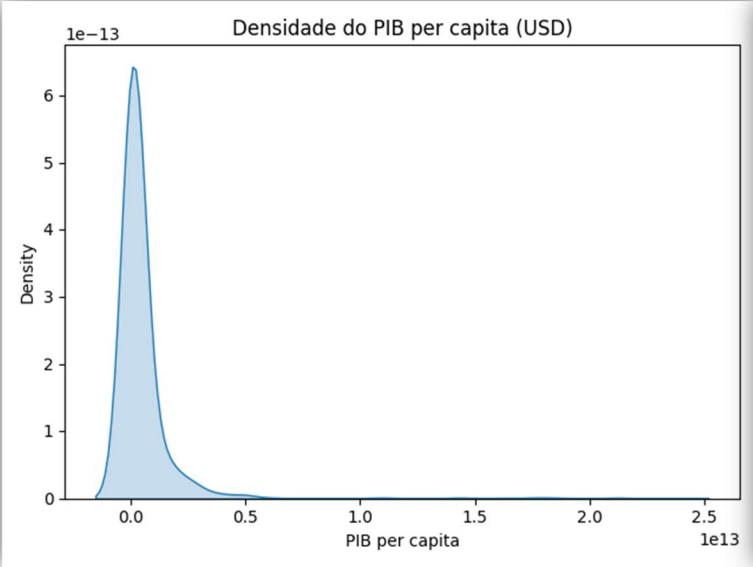


Figure 8. Kernel density estimate of GDP per capita (USD), 2011–2021. Distribution is highly right-skewed.

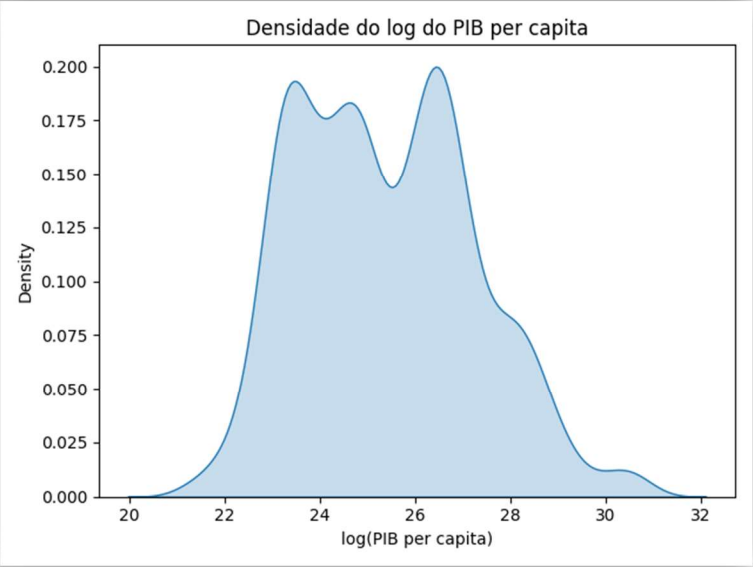


Figure 9. Kernel density of the log-transformed GDP per capita. Distribution is approximately normal and more suitable for regression analysis.

5.3 Correlation Analysis

Figure 10 displays the Pearson correlation matrix between `log_gdp` (the logarithm of GDP per capita) and the six explanatory variables considered in this study. The correlations were computed using the full panel after imputation and filtering, as implemented in the script `main_analysis.py`.

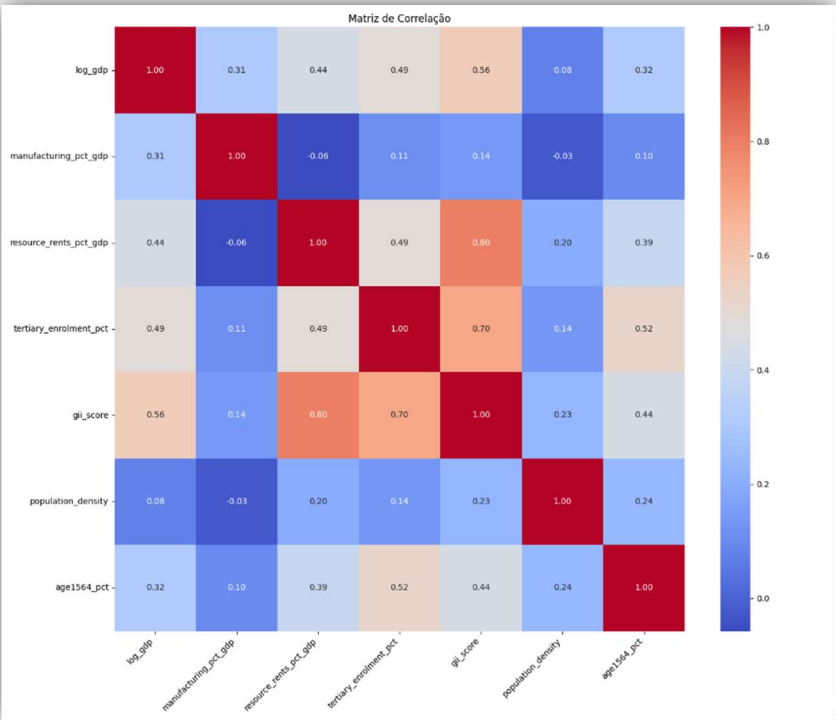


Figure 10. Pearson correlation matrix between `log_gdp` and structural predictors. Strongest correlations are observed with innovation and education variables.

Several noteworthy patterns emerge:

- Innovation (`gii_score`) shows the strongest positive correlation with income levels ($r = 0.56$), suggesting that countries with higher innovation capacity tend to have higher GDP per capita.
- Tertiary enrolment (`tertiary_enrolment_pct`) also correlates positively with `log_gdp` ($r = 0.49$), highlighting the role of higher education in economic development.
- Natural-resource rents (`resource_rents_pct_gdp`) display a moderate correlation with income ($r = 0.44$), but their role may vary depending on the country's economic structure.
- Manufacturing share (`manufacturing_pct_gdp`) presents a weaker correlation with `log_gdp` ($r = 0.31$), which may reflect heterogeneity in industrial upgrading across income groups.
- Demographic indicators show weaker links: population aged 15–64 ($r = 0.32$) and population density ($r = 0.08$) exhibit low associations with income.
- Despite the overall low-to-moderate correlations between most explanatory variables and `log_gdp`, some pairs exhibit strong interdependence. Notably, `gii_score` and `resource_rents_pct_gdp` show a Pearson correlation of 0.80, while `gii_score` and `tertiary_enrolment_pct` correlate at 0.70. These high coefficients suggest potential multicollinearity, where variables carry overlapping information.

Although multicollinearity does not compromise the model's ability to explain variance in GDP, it can lead to inflated standard errors, making individual coefficients less stable or statistically insignificant. In this project, no explanatory variables were removed because each one reflects a theoretically distinct and policy-relevant structural driver of development. Furthermore, the use of cluster-robust standard errors in regression models mitigates some of the risks associated with correlated predictors.

These insights guided the specification of regression models in later sections.

5.4 – Relationships Between Predictors and GDP

To further understand how each explanatory variable relates individually to economic output, we plotted scatterplots with regression lines for all six predictors against the logarithm of GDP per capita (`log_gdp`). These visualisations allow us to assess not only the strength and direction of each association, but also the potential presence of non-linearities or outliers.

Each plot was generated using the function `eda(df, X_cols)` from the script `main_analysis.py`, using seaborn's `regplot` for linear trend fitting. Below are the key findings from this bivariate analysis:

- Innovation (GII Score): A strong, positive, and linear relationship with `log_gdp` is evident. Countries with higher innovation scores tend to exhibit significantly higher

income levels. This is consistent with the role of technological advancement in driving productivity and competitiveness.

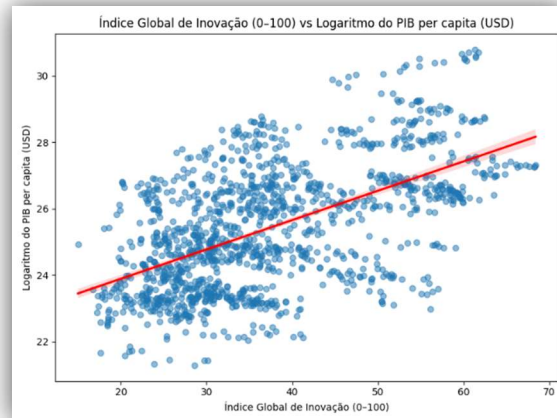


Figure 11. A strong and approximately linear relationship between innovation capacity and income. Higher GII scores are associated with higher GDP per capita.

- Tertiary Education Enrollment: The relationship is clearly positive, albeit slightly more dispersed than for innovation. Higher enrollment rates are generally associated with higher GDP per capita, highlighting the economic value of an educated workforce.

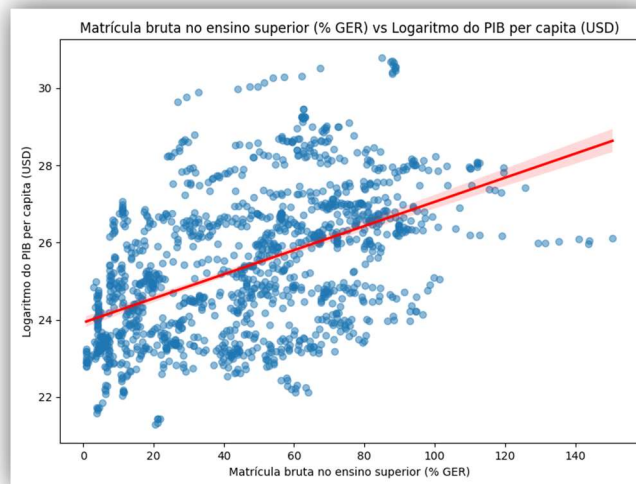


Figure 12. Positive association between tertiary education and income levels. Countries with greater access to higher education tend to have higher economic output.

- Manufacturing (% of GDP): A moderate and positive relationship is visible. While manufacturing plays a structural role in development, its explanatory power may be limited in advanced economies where services dominate.

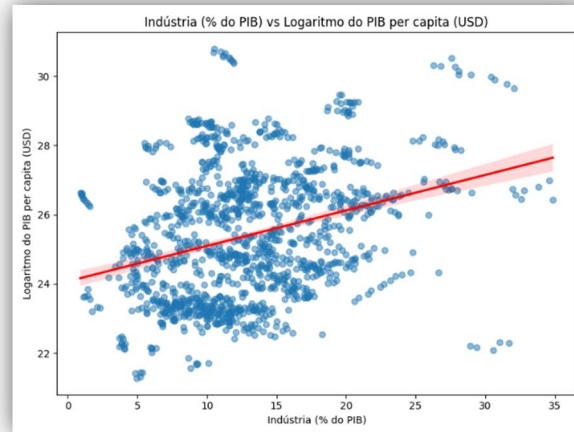


Figure 13. Moderate positive relationship, suggesting that industrialisation still plays a role in supporting higher income, though less tightly than innovation or education.

- Natural Resource Rents (% of GDP): A weak and noisy positive relationship, with many outliers. While resource-rich countries may enjoy income windfalls, the wide dispersion suggests heterogeneity in how these rents translate into broad-based economic development.

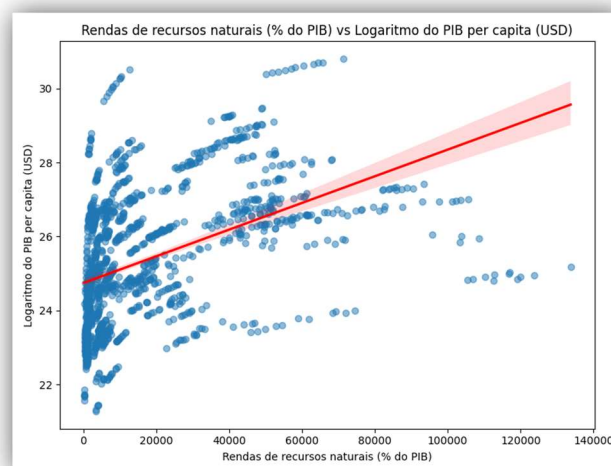


Figure 14. Weak and noisy relationship. While some resource-rich countries are wealthy, others are not—indicating that rents alone do not guarantee high GDP.

- Population Aged 15–64 (% of Total): A modest positive association exists, reinforcing the relevance of a productive-age population in contributing to output. However, the effect size is relatively small.

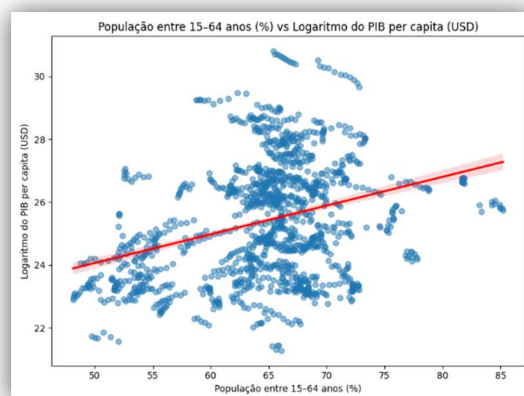


Figure 15. Mildly positive trend, showing that a larger working-age population can contribute to higher GDP, though the effect is not strong in isolation.

- Population Density (people/km²): The relationship is minimal, with considerable scatter. While dense urban areas can boost efficiency, excessive density or poor infrastructure may offset potential gains, leading to an overall weak association.

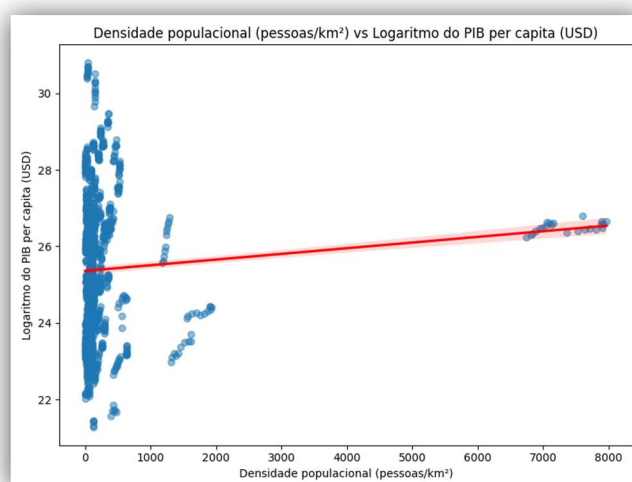


Figure 16. Very weak correlation. High density alone does not predict higher income and may be confounded by factors like urban infrastructure or inequality.

Together, these plots visually confirm and complement the correlation analysis presented earlier. They also help anticipate which variables might contribute meaningfully in multivariate models, and which ones may require careful interpretation due to weaker or non-linear effects.

5.5 Temporal Evolution in Selected Countries

Bottom 5 Countries – Key Observations

To complement the cross-sectional insights, this subsection examines the temporal dynamics of structural indicators in countries with the lowest average GDP per capita during the 2011–2021 period. The selected countries are: Burundi (BDI), Cabo Verde (CPV),

Guyana (GUY), Montenegro (MNE), and Eswatini (SWZ). The trends shown help reveal patterns of structural stagnation or gradual improvement in low-income contexts.

- Tertiary enrolment rates remain persistently low in BDI, GUY, and SWZ, with only MNE showing levels above 60%. Minimal changes over time highlight slow progress in educational access.

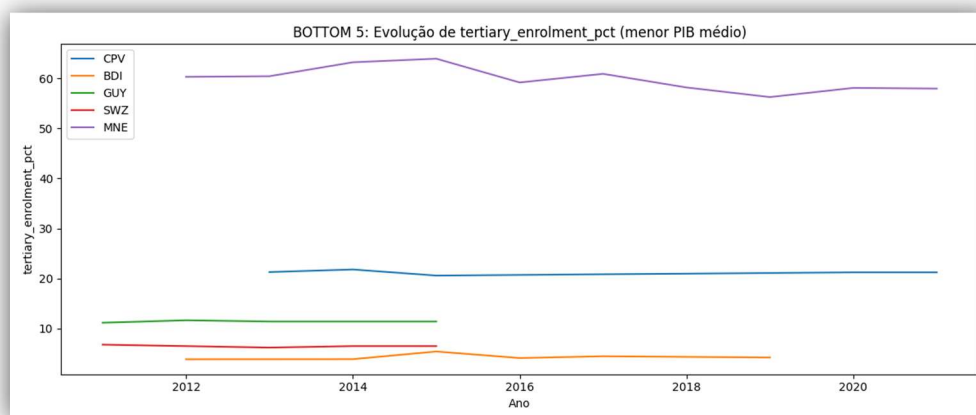


Figure 17. Evolution of tertiary enrolment rates (% gross) in the five countries with the lowest average GDP per capita (2011–2021).

- Innovation capacity is also limited, with all five countries scoring below 45 on the GII scale. Most countries exhibit flat or slightly declining trends, particularly SWZ and CPV.

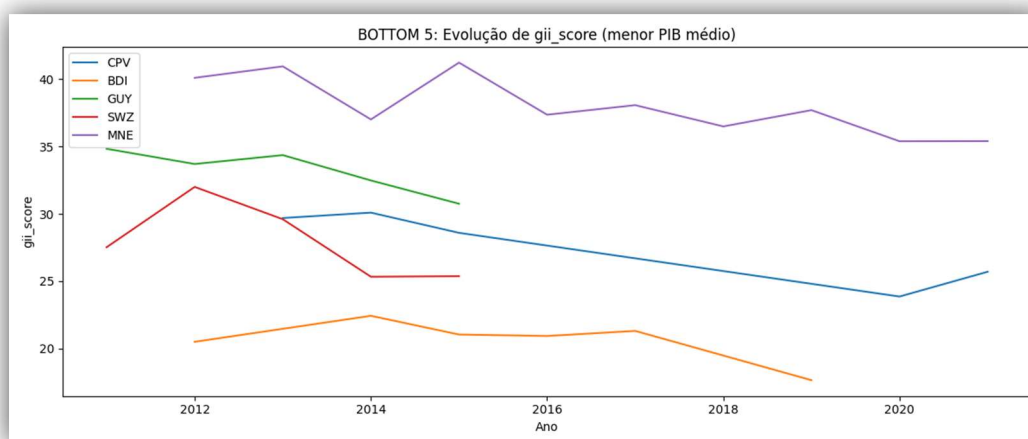


Figure 18. Evolution of Global Innovation Index scores in bottom 5 GDP countries. Most countries show low or declining innovation capacity.

- Industrial contribution to GDP is modest and relatively flat in four out of five countries, except for SWZ, which shows an industrial share above 30%—likely tied to legacy manufacturing or extractive sectors.

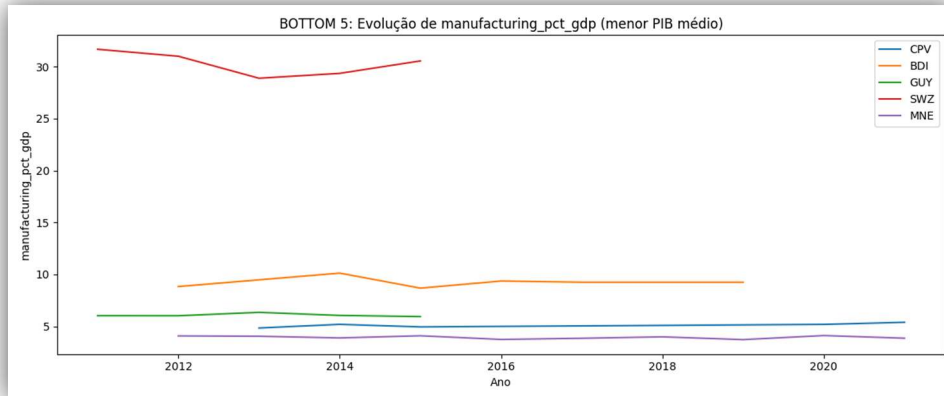


Figure 19. Manufacturing share of GDP over time in bottom 5 countries. SWZ stands out with relatively high industrial activity.

- Population density remains stable in most cases, with only BDI showing a notable upward trend.

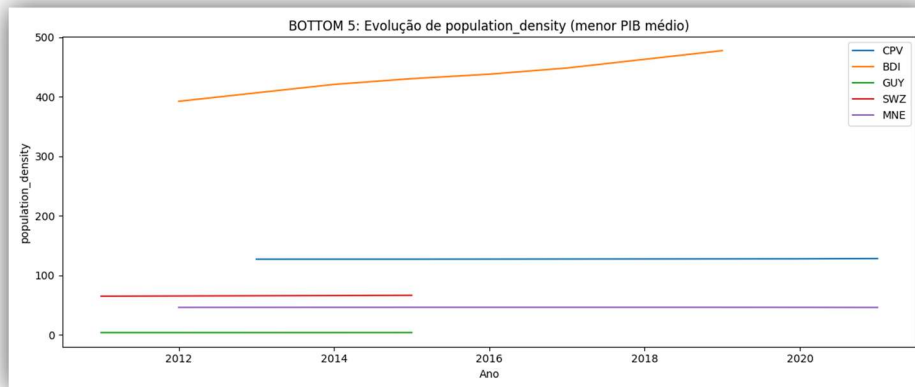


Figure 10. Population density trends in low-income countries. BDI shows a steady increase while others remain stable.

- Natural resource rents are highly heterogeneous: MNE and GUY show high values, while BDI has almost none. Resource dependency thus varies greatly within this group.

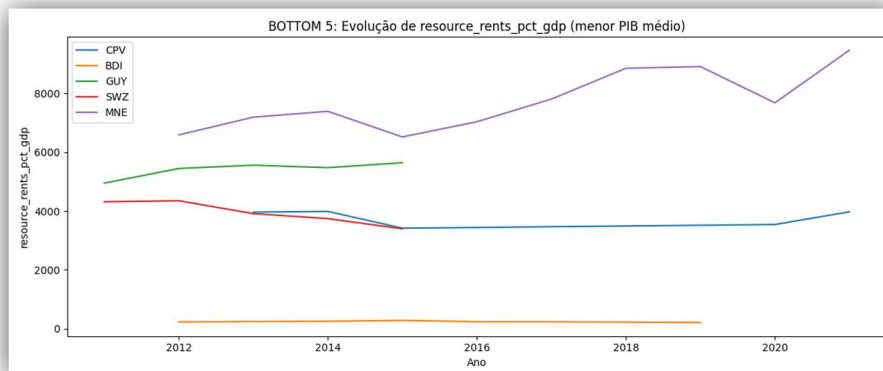


Figure 11. Natural resource rents (% of GDP) for bottom 5 countries. MNE and GUY exhibit high dependence on resource-based income.

- Working-age population share (15–64) remains fairly stable, with no sharp demographic shifts, though BDI shows slightly lower values overall.

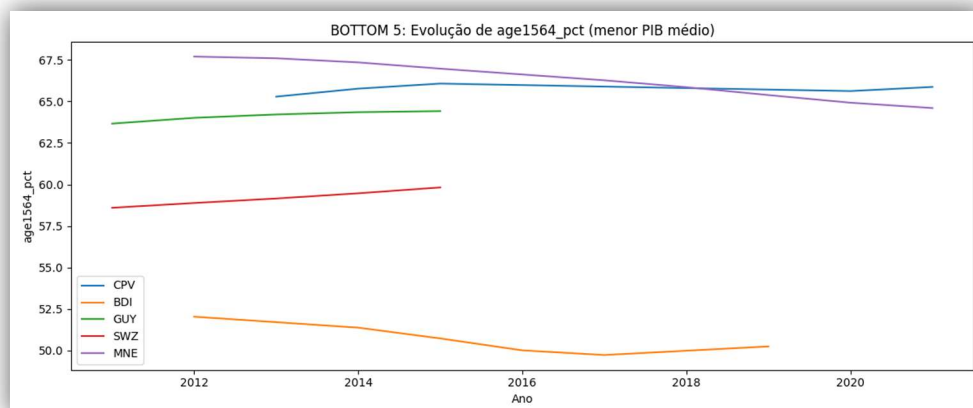


Figure 12. Share of working-age population (15–64) across bottom 5 countries, showing stable demographic structures.

These findings underscore persistent structural limitations among the poorest economies, including stagnation in educational access and innovation inputs.

Top 5 Countries – Key Observations

To complement the exploratory analysis, we examine temporal trends in the top five countries with the highest average GDP per capita between 2011 and 2021: USA, China, Japan, Germany, and United Kingdom. These countries exhibit relatively stable and high economic and innovation indicators, though with some relevant demographic and structural changes.

- Working-age population (% aged 15–64): All top countries show a gradual decline in the share of working-age population. Japan leads the decline, reflecting accelerated demographic ageing. The USA also sees a modest drop, while China remains the highest but follows a downward path.

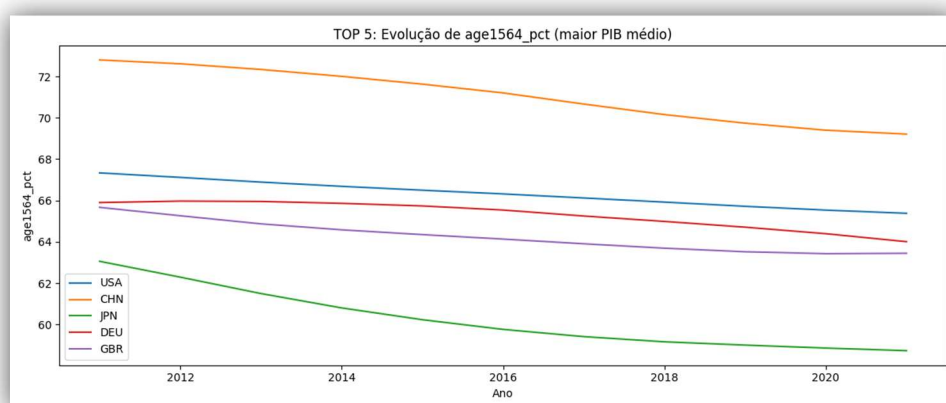


Figure 13. Evolution of the working-age population (% aged 15–64) in the top 5 countries by average GDP per capita (2011–2021). All countries show a declining trend, most notably Japan.

- Global Innovation Index (GII): These countries consistently rank high in innovation. The UK and USA maintain scores above 60, while China shows a clear upward trajectory, improving its innovation capacity throughout the decade.

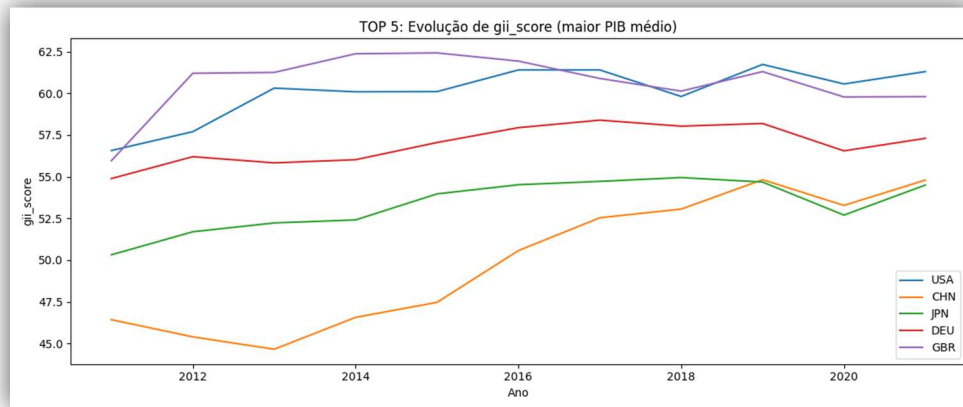


Figure 14. Evolution of GII scores (0–100) in top GDP countries. China shows the strongest upward trend, while others remain stable at high levels.

- Manufacturing (% of GDP): China leads in manufacturing contribution, though shows a decline from 32% to around 27%. Japan and Germany remain industrial powerhouses (~20%). The USA and UK show low and flat values, which may be partly influenced by imputed data in years with missing entries.

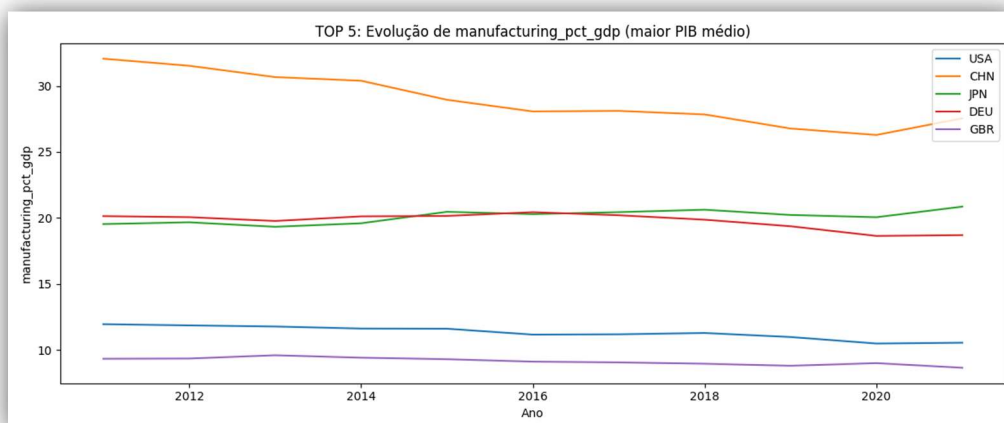


Figure 15. Evolution of manufacturing value added (% of GDP). China dominates but with a gradual decline; the USA and UK remain low.

- Population density (people/km²): Minimal change over time. Japan is the most densely populated, followed by the UK and Germany. The USA continues to have the lowest density, showing that high income does not require high population concentration.

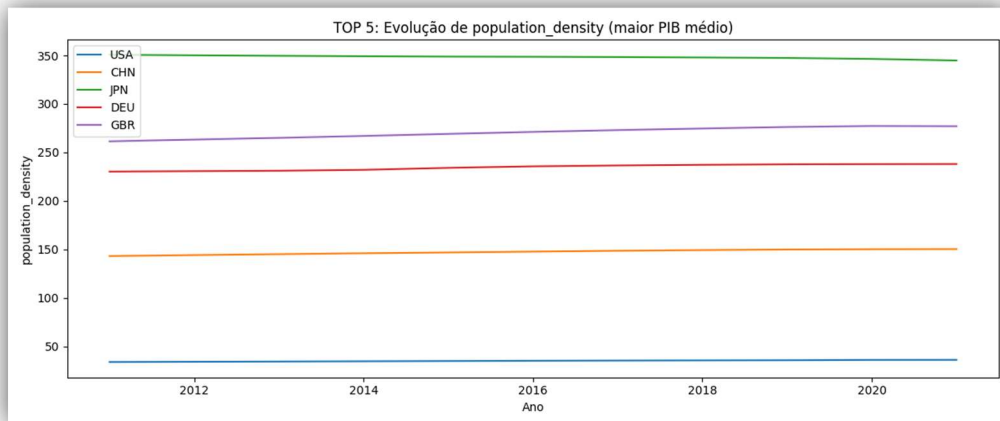


Figure 16. Population density trends in top GDP countries. Densities are stable, with wide variation across countries.

- Natural resource rents (% of GDP): The USA shows a steep increase, possibly reflecting accounting differences or rising extraction activities. Germany and the UK fluctuate moderately. Japan and China maintain consistently low rents, showing limited dependence on extractive sectors.

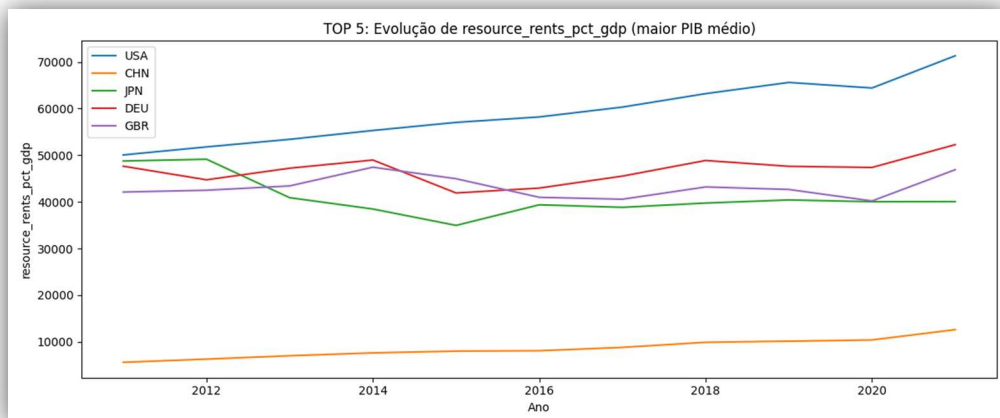


Figure 17. Trends in natural resource rents as % of GDP. The USA shows the sharpest rise, while Japan and China remain low.

- Tertiary education enrolment (% gross): The USA remains at the top (~88%). China shows remarkable progress, rising from ~27% to over 65%. Germany, Japan and the UK show steady but slower growth. Flat lines in some countries may partially reflect imputed values, though trends remain credible.

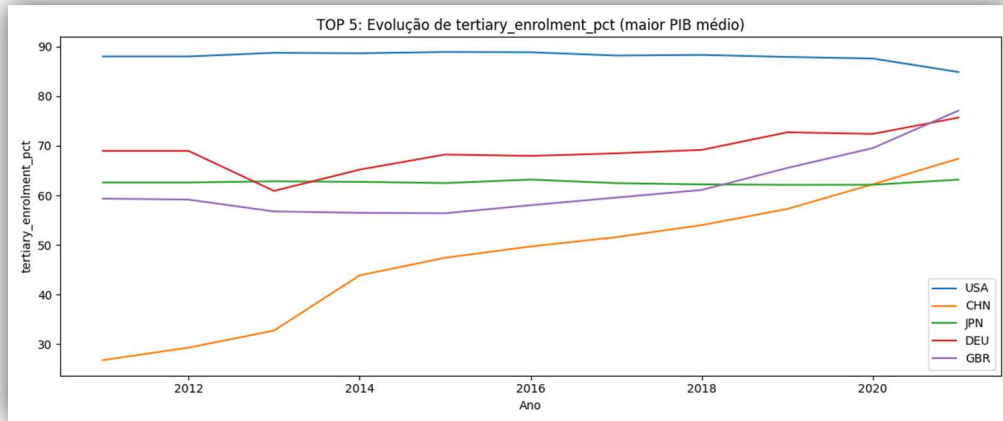


Figure 18. Trends in natural resource rents as % of GDP. The USA shows the sharpest rise, while Japan and China remain low.

6 – Methodology

To explore the structural determinants of GDP per capita across countries and years, we implemented a multi-method analytical pipeline using the final integrated dataset (panel_model_final.csv). All steps were executed in the custom script main_analysis.py.

The overall strategy combines linear regression, nonlinear modelling, and unsupervised learning. The target variable is the logarithm of GDP per capita (log_gdp), used to stabilise variance and facilitate interpretation.

Our methodology comprises the following components:

6.1 Linear Regression: OLS and Hierarchical

We start with Ordinary Least Squares (OLS) to estimate the relationship between log_gdp and six standardised structural predictors.

To account for within-country correlation over time, cluster-robust standard errors were used (grouped by country).

Additionally, we applied a hierarchical regression strategy, building three nested OLS models:

1. Model 1: Industry + Resource rents
2. Model 2: Adds Innovation
3. Model 3: Adds Education

This approach allows us to assess the incremental explanatory power of innovation and education over industrial structure.

6.2 Random Forest Regression

As a nonlinear and non-parametric alternative, we trained a Random Forest Regressor on the same six predictors. This model is robust to multicollinearity and captures potential nonlinear relationships.

We report R^2 scores and feature importance metrics using 5-fold cross-validation, enabling a comparative view against the OLS approach.

6.3 K-Means Clustering

To investigate the possibility of grouping countries by structural economic profiles, we applied K-Means clustering on the country-level means of all explanatory variables. Prior to clustering, features were normalised (z-score).

The optimal number of clusters was chosen based on the elbow method, and the resulting clusters were visualised using radar plots.

7 – Modelling Results

This section presents the results of the statistical modelling process, conducted to assess the structural determinants of GDP per capita across countries. Three modelling approaches were employed: Ordinary Least Squares (OLS) regression, hierarchical regression models, and Random Forests. Each of these methods provides complementary insights into the relationships between GDP and explanatory variables such as innovation, industrial structure, education, and demography.

The objective is not only to identify the most relevant predictors but also to compare the explanatory power and consistency of the results across modelling strategies. All models were estimated using the fully-imputed panel dataset described in previous sections (`panel_model_final.csv`), focusing on the log-transformed GDP per capita (`log_gdp`) as the dependent variable.

The following subsections present the results of each modelling approach in turn, highlighting key coefficients, measures of model fit, and variable importance, as well as noting statistically significant patterns that will later be discussed in the context of the research questions.

7.1 OLS Regression Results

To explore how structural factors relate to income levels across countries, we estimated a linear regression model where the dependent variable is the logarithm of GDP per capita. The model uses all six explanatory variables, standardized beforehand to allow meaningful coefficient comparisons.

The method used was Ordinary Least Squares (OLS) with clustered standard errors by country, which accounts for repeated observations across years for the same country.

The results, summarized in `terminal_output.txt` that goes with this report, show that:

- The overall model explains about 40% of the variation in log GDP per capita across countries and years.
- Among the six predictors, three have statistically significant effects at the 5% level:
 - Innovation (GII score) has the strongest positive effect on income levels.
 - Higher education (tertiary enrolment) also shows a significant positive relationship.
 - Manufacturing (% of GDP) is positively associated with income and statistically relevant.
- The other variables – resource rents, population density, and working-age population – do not show significant effects in this model.

The relative magnitude and direction of each variable's influence are illustrated in Figure below, which plots the standardized coefficients and their confidence intervals.

These findings suggest that innovation capacity and educational attainment are more closely associated with higher income levels than traditional factors such as natural resource abundance or population size.

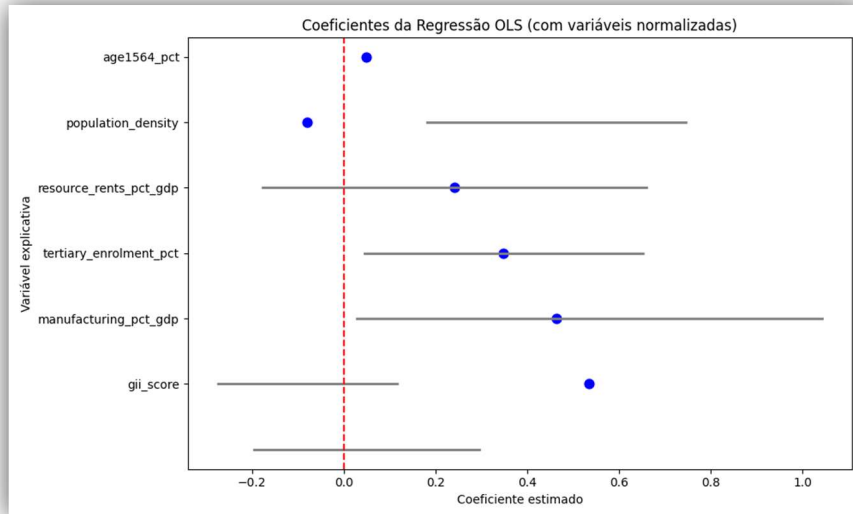


Figure 19. Coefficient Plot for OLS Regression (Standardized Predictors).

7.2 Hierarchical Regression Models

To assess the incremental explanatory power of innovation and higher education over industrial structure and resource dependence, we estimated three hierarchical Ordinary Least Squares (OLS) regression models. All models include standardized predictors and cluster-robust standard errors by country.

- **Model 1 – Industry + Resources:** This baseline model includes only `manufacturing_pct_gdp` and `resource_rents_pct_gdp`. Both predictors show positive and statistically significant coefficients ($p < 0.001$), indicating a direct association with GDP per capita. The model explains approximately 30.8% of the variance in log GDP per capita.

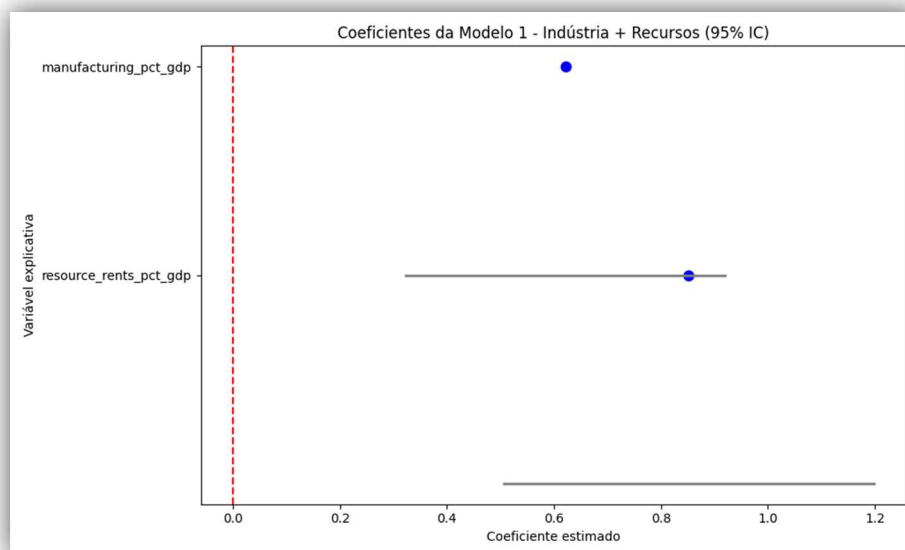


Figure 20. Estimated coefficients and 95% confidence intervals for Model 1 (Industry + Resources). Both predictors are positively and significantly associated with log GDP.

- Model 2 – + Innovation: Adding `gii_score` increases the R^2 to 37.4%. Innovation shows a strong and statistically significant effect (coef ≈ 0.83 , $p < 0.001$), even when controlling for industry and resource dependence. Interestingly, `resource_rents_pct_gdp` loses significance, suggesting its effect may be partially mediated by innovation capacity.

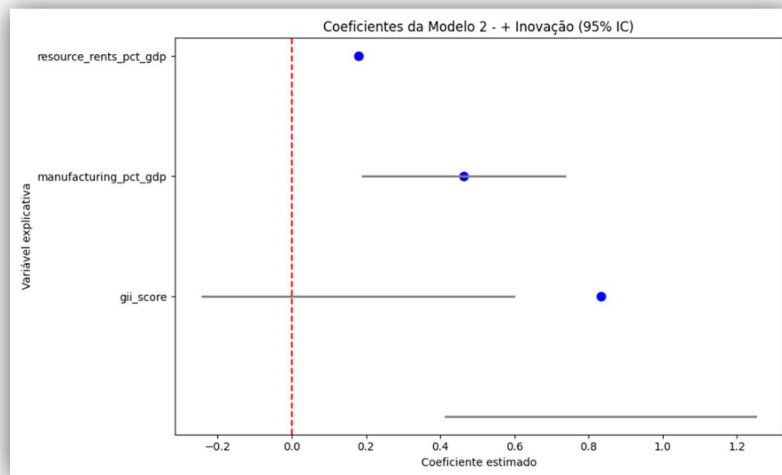


Figure 21. Estimated coefficients and 95% confidence intervals for Model 1 (Industry + Resources). Both predictors are positively and significantly associated with log GDP.

- Model 3 – + Innovation + Education: Including `tertiary_enrolment_pct` further increases explanatory power ($R^2 = 39.5\%$). Higher education enrollment is positively and significantly associated with log GDP ($p = 0.010$), reinforcing its relevance as an independent economic driver. Both innovation and manufacturing remain significant, while resource rents continue to show no strong effect.

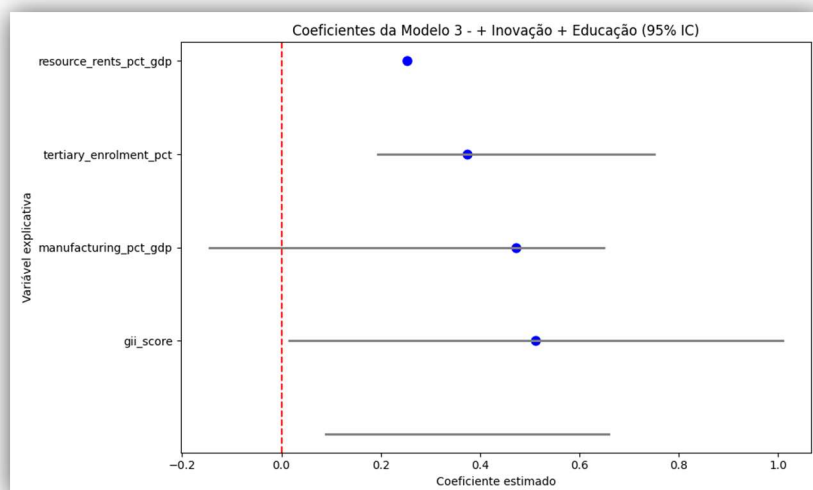


Figure 32. Estimated coefficients and 95% confidence intervals for Model 3 (+ Innovation + Education). Education adds further explanatory value to the model.

7.3 Random Forest Regression

To complement the linear models previously explored, we trained a Random Forest regressor using the same six explanatory variables. The objective of this approach was to capture non-linear relationships and interactions between variables that may not be adequately represented in linear models such as OLS.

Random Forest is a tree-based ensemble method known for its flexibility and robustness, particularly in high-dimensional or complex datasets. It can model non-linear effects without requiring prior transformation or specification of the functional form.

Model Performance:

- R^2 on training set: 0.991 This indicates that the model fits the training data extremely well, capturing nearly all the variance in the log-transformed GDP per capita.
- R^2 (cross-validation average): 0.307 When evaluated through 5-fold cross-validation — a method that simulates unseen data — the performance dropped significantly. This suggests that the model may be overfitting, memorizing patterns specific to the training data rather than generalizing to new observations.

Variable Importance:

The bar plot below displays the relative importance of each predictor variable in the Random Forest model. These scores reflect how often and how effectively each variable contributed to reducing prediction error across the ensemble of trees.

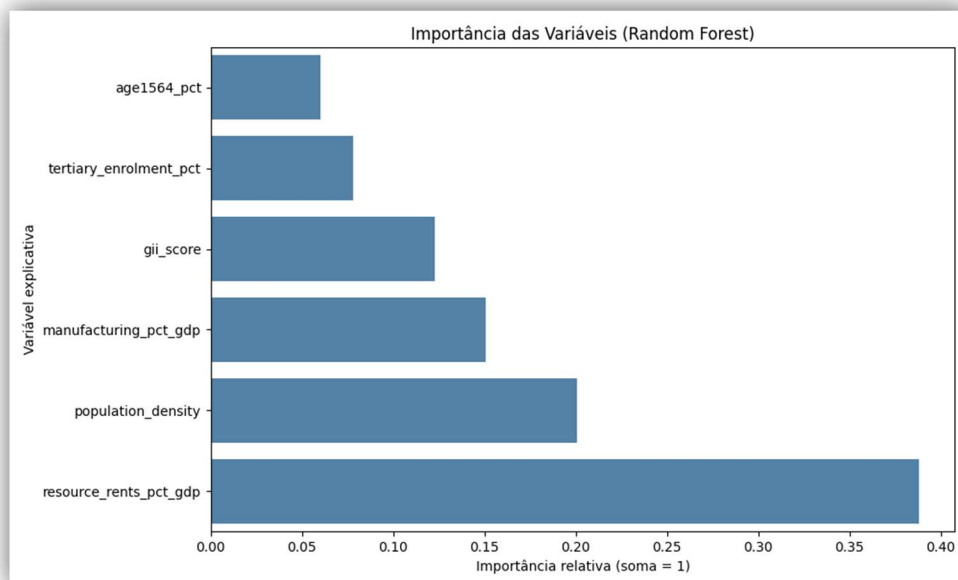


Figure 33. Variable Importance in Random Forest Model.

Key observations:

- Resource rents (% of GDP) and population density were the most influential variables in predicting GDP per capita in this model.
- Manufacturing output and the Global Innovation Index also contributed meaningfully.

- Higher education enrollment had lower importance, despite showing significance in the OLS models.
- Age structure (15–64 years) received the lowest importance score, consistent with its weak coefficient in the OLS model.

These results suggest that Random Forest, while powerful, emphasizes variables with stronger and more complex effects. However, its modest cross-validation performance indicates limited generalizability compared to more interpretable models like OLS.

Interpretation Note on Variable Discrepancy:

It is worth highlighting a notable discrepancy between the OLS and Random Forest results regarding the variable *resource_rents_pct_gdp*. In the OLS regression, this variable showed a positive coefficient but was not statistically significant ($p \approx 0.26$), suggesting no strong or consistent linear relationship with GDP per capita across the dataset. However, in the Random Forest model, it emerged as the most important variable, accounting for nearly 39% of total predictive importance.

This divergence reflects fundamental differences in how both models operate. OLS assumes a constant and linear effect across all observations, while Random Forest captures non-linear patterns and interactions among variables. This suggests that *resource_rents* may influence GDP in more complex, context-specific ways — such as differing effects across resource-rich versus resource-poor countries — which are not well captured by a simple linear model.

Such contrast reinforces the value of combining multiple modelling approaches to gain a more nuanced understanding of the structural drivers of economic performance

7.4 Clustering Analysis

To complement the regression and machine learning models, we applied an unsupervised clustering technique using the K-Means algorithm to identify groups of countries with similar structural profiles. The objective was to explore whether countries can be meaningfully grouped based on common patterns in the key explanatory variables, and to interpret these clusters in terms of economic structure and development.

We first computed the average value of each structural variable across the 2011–2021 period for each country. The selected variables were: manufacturing as a share of GDP, resource rents as a share of GDP, Global Innovation Index (GII), tertiary enrolment rate, population density, and the share of working-age population (15–64 years). These variables were then standardized using z-score normalization, meaning that all variables were rescaled to have mean 0 and standard deviation 1. This step ensures comparability across variables with different units and scales, and is crucial for distance-based methods such as KMeans.

The optimal number of clusters ($k=3$) was determined using the elbow method, where the reduction in distortion begins to level off after three clusters.

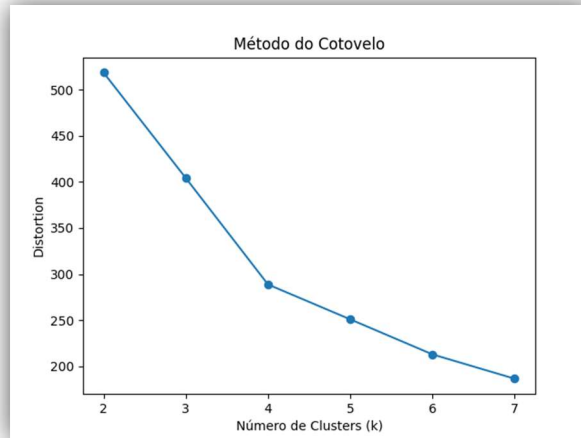


Figure 34. Elbow method to determine the number of clusters. The distortion curve suggests $k = 3$ as a suitable choice.

Cluster Characteristics:

The figures present the radar plots for the three clusters, which can be interpreted as distinct economic profiles:

Cluster 0 represents countries with relatively balanced profiles. These countries exhibit moderately high levels of industrialization and a high proportion of the population in the working-age bracket (15–64 years), suggesting a favourable demographic structure for economic activity. Their average levels of tertiary enrolment and innovation are around the global median, indicating access to education and moderate investment in knowledge and technology. Importantly, countries in this group have minimal reliance on natural resource rents, implying a more diversified or self-sustaining economy. Overall, Cluster 0 could be interpreted as comprising emerging or stable mid-level economies with demographic and industrial advantages but moderate performance in innovation and education.

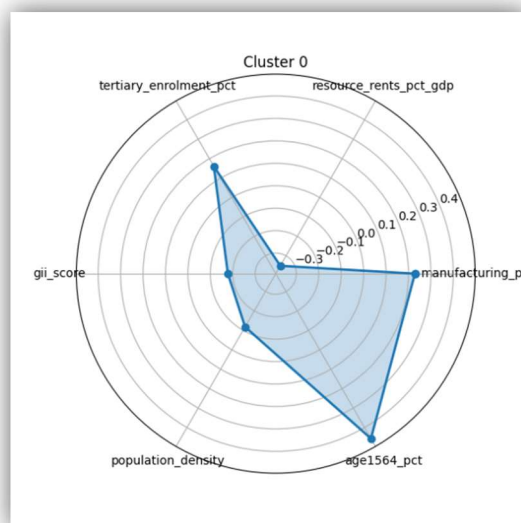


Figure 35. Radar plot for Cluster 0 (standardized values). Countries with above-average manufacturing and age structure, moderate education, and below-average innovation and resource reliance.

Cluster 1, by contrast, includes countries with high values in nearly all dimensions — especially resource rents, innovation (GII), and tertiary education enrolment. These countries also display high population density and reasonable age structure, although their manufacturing share is closer to the global average. This profile suggests countries that are both resource-rich and capable of converting that wealth into investments in human capital and institutional development. Some of these may be upper-middle-income economies or high-income countries with strong extractive sectors and advanced innovation systems. The combination of high GII and tertiary enrolment in this cluster is particularly notable, indicating structural capacity for sustained long-term growth.

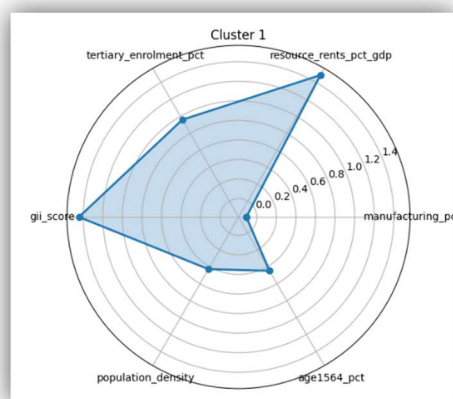


Figure 36. Radar plot for Cluster 1 (standardized values). Countries with strong innovation, high resource rents, high tertiary enrolment, and above-average population density — structurally well-positioned.

Cluster 2 captures structurally disadvantaged economies. These countries have negative standardized values across all six dimensions, meaning they fall below the global average in each indicator. In particular, they show very low levels of tertiary education, innovation, and working-age population share. Their dependence on resource rents and industrial share of GDP is also limited, and population density is comparatively low. This profile suggests economies facing compound challenges — lacking both demographic dividends and institutional capacity — which may hinder their development prospects. Countries in this cluster are likely to be low-income, with persistent barriers in both human capital and structural diversification.

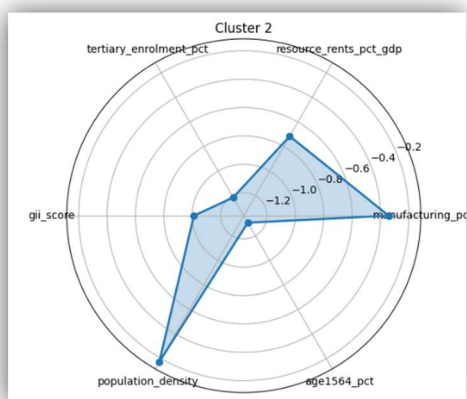


Figure 37. Radar plot for Cluster 2 (standardized values). Countries with below-average performance in all dimensions — reflecting structural constraints and low development capacity.

In summary, clustering analysis reveals three distinct country profiles that cut across geographic and income boundaries. By working with normalized data, the radar plots offer a clear view of how countries compare relative to global averages, rather than in absolute terms. This approach emphasizes patterns and divergences in structural composition, helping to identify clusters of countries that may benefit from similar policy interventions.

8 - Discussion

This chapter addresses the central research questions that guided the project, interpreting the modelling outcomes presented in the previous section. The goal is to draw substantive conclusions about which structural drivers matter most for GDP per capita, whether knowledge-based factors outperform resource-based ones, and whether countries can be meaningfully grouped based on these drivers.

Rather than simply repeating statistical outputs, this section synthesizes findings across different modelling techniques — including linear regression (OLS), hierarchical regression, Random Forest and clustering — in order to draw coherent insights. For each question, we indicate which results support the conclusion and how the different analytical methods complement (or contradict) each other.

Although exploratory analysis offered initial insights into the distributions, correlations and bivariate relationships among variables, its role was primarily descriptive. Therefore, it is not the focus here. Instead, this discussion centers on inferential and predictive modelling results, which are more robust for answering causal or explanatory questions and for testing hypotheses on economic structure.

Below, we revisit the three core guiding questions, followed by additional secondary questions that emerged throughout the modelling phase.

1. Which structural factors explain most of the cross-country variance in GDP per capita?

The multivariate OLS regression ($R^2 = 0.397$) indicates that innovation (GII score), higher education (tertiary enrolment), and manufacturing (% of GDP) are positively associated with GDP per capita, even after controlling for other structural variables. Among these, the GI score showed a statistically significant effect ($p < 0.05$), as did tertiary enrolment and manufacturing — all with moderate coefficients and confidence intervals that did not cross zero.

In Random Forest results, the most important variable in predicting log GDP per capita was resource rents (% of GDP), followed by population density and manufacturing. This indicates that non-linear relationships may amplify the effect of resource dependence, which wasn't significant in the linear model but gained weight in the non-parametric setting.

Thus, innovation, education, industry and natural resources emerge as the key explanatory drivers — though their importance varies depending on the modelling technique.

2. Do innovation and higher education outweigh natural resources once industrial structure is controlled for?

The hierarchical regression models directly address this. Starting with Model 1 (industry + resources), both variables were significant ($R^2 = 0.308$). When adding innovation (Model 2), the R^2 increased to 0.374, and the GI coefficient was highly significant ($p <$

0.001). In Model 3, the inclusion of tertiary enrolment raised R^2 to 0.395, and both innovation and education remained significant.

This progression shows that innovation and higher education do contribute additional explanatory power beyond natural resources and industrial output. In fact, resource rents lost significance in Model 3 ($p = 0.213$), suggesting that their role diminishes when knowledge-based variables are included.

3. Can countries be clustered into distinct economic profiles based on these drivers?

Yes. Using KMeans clustering ($k=3$), countries were grouped into:

- Cluster 0: High manufacturing, low resource rents, high working-age share. This cluster aligns with structurally industrial economies.
- Cluster 1: High innovation and resource rents, moderate education. Possibly resource-rich innovation leaders.
- Cluster 2: Low across most drivers, particularly in education, innovation, and working-age share — suggesting lagging development profiles.

Radar plots and normalized values confirm that structural patterns group countries into clearly distinguishable economic profiles, aligned with differing combinations of demographic, educational, industrial and innovation factors.

Secondary Questions:

- What is the relative importance of innovation, education, and resource rents when assessed jointly?

In OLS, innovation (GII) had the strongest effect ($\beta \approx 0.53$), followed by education (0.35) and manufacturing (0.46). Resource rents were not significant. In Random Forest, resource rents dominated the feature importance ($\approx 39\%$). This indicates that in non-linear models, resource wealth exerts a stronger predictive influence, likely due to threshold effects or interactions not captured in linear regression.

- Do structural drivers have different effects in low-income versus high-income contexts?

Temporal evolution plots suggest differing dynamics. In top GDP countries, tertiary enrolment is consistently high and increasing. In contrast, in bottom-income countries, enrolment and GII are low and often stagnant. Thus, structural drivers may have different baselines and marginal effects depending on the development level.

- Is there evidence of interaction or substitution between innovation and education?

While not directly modelled through interaction terms, the inclusion of both variables in Model 3 shows that they remain jointly significant. Their correlation is moderate ($r \approx 0.70$), suggesting some overlap but also complementary effects. This warrants future exploration through interaction terms or structural equation modelling.

- How relevant are demographic and spatial factors such as population density or working-age share?

In both OLS and Random Forest, these factors had low to moderate coefficients. Age share was weakly significant in OLS, and of low importance in Random Forest. However, clustering analysis showed that clusters differ notably in these dimensions, indicating that they may play a contextual rather than direct explanatory role.

- Do non-linear models like Random Forest outperform traditional regression in predictive power?

No. Although Random Forest achieved a very high training R^2 (0.991), its cross-validation R^2 was only 0.307 — lower than the OLS model's cross-val R^2 (~0.39). This indicates overfitting: the model fits the training data too closely and fails to generalize well.

The likely reasons are:

- The sample size (~1350) limits the benefit of complex models.
- The structural relationships in the data are largely linear.
- The Random Forest model captures noise along with signals, reducing out-of-sample performance.

In contrast, OLS is more stable and generalizable in this context. Despite being simpler, it provided better predictive performance and should be preferred for inference.

9 – Conclusion

This project set out to investigate the structural determinants of GDP per capita across more than 128 countries over the 2011–2021 period. Through a combination of regression models, machine learning, and clustering techniques, we aimed to identify which economic and social drivers best explain income differences between countries, and whether distinct development profiles can be uncovered.

The analysis revealed that industrial structure, innovation, and higher education are all positively associated with economic performance. In particular, the share of manufacturing in GDP and the Global Innovation Index score consistently emerged as strong predictors. Hierarchical regression showed that innovation and tertiary enrolment add explanatory power beyond natural resource rents, suggesting that intangible drivers like knowledge and education may play a more sustainable role in long-term development.

While the Random Forest model achieved near-perfect fit on the training data, it underperformed in cross-validation, pointing to overfitting and reinforcing the robustness of linear models in this context. The clustering analysis further highlighted that countries can be grouped into three distinct profiles, differentiated primarily by their industrial intensity, resource dependency, and demographic composition.

Overall, the results support the hypothesis that knowledge-based and diversified economies tend to achieve higher income levels, while those heavily reliant on resources may face structural limitations. These findings may inform future policy debates around economic diversification, education investment, and innovation promotion.

10 - AI Contribution

Artificial Intelligence tools played a supportive but meaningful role in the completion of this project. Specifically:

- Natural language models were used to generate code templates and assist in the debugging of data processing scripts (e.g., for regression analysis and clustering).
- AI was consulted to refine interpretations of statistical outputs, helping to clarify complex findings such as the role of interaction effects or the implications of overfitting in non-linear models.
- In the reporting phase, generative models were useful to restructure technical content into accessible academic language, enhancing both clarity and coherence.
- Visual design and explanatory captions for plots were also iteratively improved with the help of AI suggestions, ensuring better communication of results.

All decisions regarding modelling, interpretation, and analysis were made critically by the author. AI was used as an assistant — not a substitute for analytical thinking or domain understanding.

10 – References

- [1] The World Bank, “DataBank | The World Bank,” *Worldbank.org*, 2025.
<https://databank.worldbank.org/>
- [1] K. Kovacs, “Global Innovation Index (WIPO) - 2011-2024 Data,” *Kaggle.com*, 2024.
<https://www.kaggle.com/datasets/karlakovacs/global-innovation-index-wipo-2011-2024-data>

11 – Appendix

This appendix provides an overview of the files submitted alongside this report and the folder structure used to organize the datasets, scripts, and outputs referenced.

Delivery_P2-SGD/Docs_P2/Analysis/

Contains the main scripts, outputs, and visualizations from the final analysis phase:

- main_analysis.py – Main pipeline script that executes data loading, preprocessing, OLS regressions, Random Forest modelling, clustering, and plot generation.
- panel_model_final.csv – Final dataset used in modelling, after filtering and imputation.
- terminal_output.txt – Console log capturing printed model summaries, scores, and diagnostics.
- /plots/ – Folder with all figures referenced in the report (e.g., regression coefficients, clustering plots, distributions) and supporting .csv files:

Delivery_P2-SGD/Docs_P2/Datasets_extracted/

Includes the original raw indicators extracted from sources such as the World Bank and GII portal:

- GII_score/
- NV.IND.MANF.ZS_manufacturing_pct_gdp/
- NY.GDP.TOTL.RT.ZS_resource_rents_pct_gdp/
- SE.TER.ENRR_tertiary_enrolment_pct/
- SP.POP.1564.TO.ZS_age1564_pct/
- EN.POP.DNST_population_density/
- NY.GDP.PCAP.CD_gdp_per_capita_current_usd/

Delivery_P2-SGD/Docs_P2/Datasets_extracted/

Includes the original raw indicators extracted from sources such as the World Bank and GII portal:

- GII_score/
- NV.IND.MANF.ZS_manufacturing_pct_gdp/
- NY.GDP.TOTL.RT.ZS_resource_rents_pct_gdp/
- SE.TER.ENRR_tertiary_enrolment_pct/
- SP.POP.1564.TO.ZS_age1564_pct/
- EN.POP.DNST_population_density/
- NY.GDP.PCAP.CD_gdp_per_capita_current_usd/

Delivery_P2-SGD/Docs_P2/Datasets_merged/

Includes the datasets resulting from SQL processing, merging, and imputation:

- /Datasets_returned_pgadmin/
 - panel_model.csv
 - panel_full.csv
 - /panel_model_after_imputation/ – Contains script imputing missing values.
- /Sql_queries/ – SQL scripts used to construct, join, and filter data tables within PostgreSQL (pgAdmin workflow).